RL & DP Notes

1. Dynamic Programming (DP)

Core Idea: Solve Bellman equations using a known model.

$$V(s) = \sum_{s'} P(s'|s,a)\big[r + \gamma V(s')\big]$$

- Requires full transition model.
- Uses expectation over all next states.
- Deterministic updates.
- Bootstraps using $V(s')$.

Key Property: Exact expectation backup.

2. Monte Carlo (MC)

Core Idea: Learn from complete sampled episodes.

Return:

$$G_t = \sum_{k=0}^{T-t} \gamma^k r_{t+k+1}$$

Update:

$$V(s) \leftarrow V(s) + \alpha(G_t - V(s))$$

- No model required.
- No bootstrapping.
- High variance, low bias.
- Update only after episode ends.

Key Property: Sample-based full-return backup.

3. Temporal Difference (TD)

Core Idea: Sample-based bootstrapping.

TD(0) update:

$$V(s) \leftarrow V(s) + \alpha\big[r + \gamma V(s') - V(s)\big]$$

TD error:

$$\delta = r + \gamma V(s') - V(s)$$

- No model required.
- Bootstraps.
- Lower variance than MC.
- Sample-based approximation of DP.

Key Difference:

- DP = expectation backup
- TD = sample backup

4. Value-Based vs Policy-Based RL

Value-Based (e.g., Q-learning)

$$Q(s,a) \leftarrow r + \gamma \max_{a'} Q(s',a')$$

- Learn value function.
- Derive policy via arg max.
- Off-policy (greedy target, exploratory behavior).

Policy Gradient (PG)

$$\nabla_\theta J(\theta) = \mathbb{E}\big[\nabla_\theta \log \pi_\theta(a|s) A(s,a)\big]$$

- Directly optimize policy.
- No arg max derivation.
- Advantage reduces variance.

5. Advantage & GAE

Advantage:

$$A(s,a) = Q(s,a) - V(s)$$

TD error:

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$$

GAE:

$$A_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}$$

- $\lambda = 0 \rightarrow$ TD (low variance, high bias)
- $\lambda = 1 \rightarrow$ MC (low bias, high variance)

GAE interpolates between TD and MC.

6. PPO (Proximal Policy Optimization)

Ratio:

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\text{old}}(a_t|s_t)}$$

Clipped objective:

$$L = \min\Big(r_t A_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon)A_t\Big)$$

- On-policy.
- Actor-Critic.
- Uses GAE for advantages.
- Clipping approximates trust region.
- KL monitors policy shift.

Core Idea: Improve policy without large destructive updates.

7. DP vs TD vs RL vs ADP

Method	Model	Sampling	Bootstrapping
DP	Yes	No	Yes
MC	No	Yes	No
TD	No	Yes	Yes

Insight: RL can be viewed as data-driven Approximate Dynamic Programming (ADP).

8. DDP (Differential Dynamic Programming)

Core Idea: Second-order trajectory optimization.

- Requires differentiable model.
- Uses local quadratic expansions.
- Solves finite-horizon optimal control.
- Often used inside MPC.

If model is differentiable:

$$\nabla_\theta J = \text{Backprop through dynamics}$$

DDP operates in model-based optimal control regime.

9. Conceptual Unification

All methods originate from the Bellman principle.

- DP: exact Bellman solution.
- TD: stochastic Bellman approximation.
- RL: model-free Bellman learning.
- PG: direct optimization of expected return.
- PPO: stable policy gradient with trust control.
- DDP: differentiable optimal control via second-order methods.

Grand View:

$$\text{Control} \rightarrow \text{Dynamic Programming} \rightarrow \text{Approximation} \rightarrow \text{Reinforcement Learning}$$